

# NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

## Fifth Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

### 20ME503 / 17ME503 - DYNAMICS OF MACHINES

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

#### Unit – I

Marks BT\* CO\* PO\*

1. a) Explain the equilibrium with respect to  
i) 2- force ii) 3- force iii) 2-force & a Torque
- b) Determine the input torque (T) on the crank of a slider-crank mechanism shown in Fig. 1(b).

6 L\*1 1 1

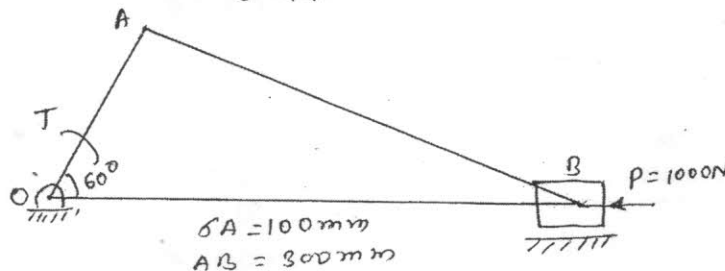


Fig. 1(b)

14 L3 1 2

2. a) Explain 'D' Alembert's principle.
- b) The following data relate to a horizontal reciprocating engine shown in Fig. 2(b). Mass of reciprocating parts = 120kg, Crank length = 90mm, Engine speed = 600 rpm. Connecting rod mass 90kg length between centers = 450mm, distance of center of mass from big end centre = 180mm. Radius of gyration about an axis through centre of mass 150mm. Find the magnitude and the direction of the inertia torque on the crank shaft when the crank turned 30° from inner dead centre.

6 L1 1 1

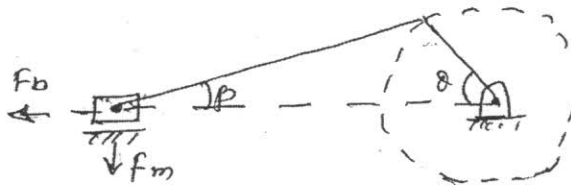


Fig. 2(b)

14 L3 1 2

#### Unit – II

3. a) Define i) sensitiveness ii) Isochronism iii) controlling force iv) stability v) effect and power of a governor.
- b) The mass of each ball of a Hartnell Governor is 1.4kg. The length of ball arm of the bell crank lever is 100mm, whereas the length of arm towards sleeve is 50mm. The distance of fulcrum of bell crank lever from the axis of rotation is 80mm. The extreme radii of rotation of the balls are 75mm and 112.5mm. The maximum speed is 6% greater than the minimum equilibrium speed which is 300 rpm. Determine i) stiffness of the spring ii) equilibrium speed when radius of rotation of ball is 90mm, neglect obliquity of the arms.
4. a) Derive an expression for Gyroscopic couple.
- b) Explain the Gyroscopic effect of steering pitching and rolling of a naval ship in sea.

10 L1 3 1

10 L3 3 2

6 L1 3 1

6 L2 3 1

- c) An Aeroplane makes a complete half circle of 50m radius towards left when flying at 200 km/Hr. The mass of rotary engine and propeller is 400kg with radius of gyration 300mm. The engine runs at 3000 rpm counterclockwise when viewed from rear. Determine gyroscopic couple and its effect on the Aircraft.

8 L3 3 2

## Unit - III

5. a) Discuss on the concept of static and dynamic balancing.  
b) Four masses A, B, C and D are completely balanced. Masses C & D make angle of  $90^\circ$  and  $210^\circ$  respectively with B in the same sense. The planes containing B & C are 300mm apart, masses A, B, C & D can be assumed to be concentrated at radii of 360mm, 480mm, 240mm and 300mm respectively. The masses B, C and D are 15kg, 25kg and 20kg respectively. Determine  
i) mass A and its angular position ii) position of planes A & D.
6. A five cylinder in-line engine running at 750rpm has successive crank  $144^\circ$  apart, the distance between the cylinder centre lines being 375mm. The piston stroke is 225mm and the ratio of the connecting rod to the crank is 4. Examine the engine for balance of primary and secondary forces and couples. Find the maximum values of these and the position of central crank at which maximum value occurs. The reciprocating mass for each cylinder is 15kg.

4 L1 3 1

16 L3 3 2

20 L3 2 2

## Unit - IV

7. a) Define the following.  
i) SHM ii) Degrees of freedom iii) Frequency iv) Damped and undamped vibration v) Amplitude vi) Free and forced vibration.  
b) A harmonic motion is given by the equation  $x = 6 \sin(2t + \phi)$ . Determine its two components such that one leads it by  $30^\circ$  and other lags it by  $60^\circ$ .  
c) Find the Natural frequency of the system give in Fig. 7(c).

8 L1 4 1

6 L2 4 2

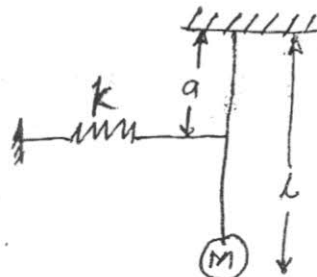


Fig. 7(c)

6 L3 4 2

8. a) Find the natural Frequency of the system shown in Fig. 8(a).

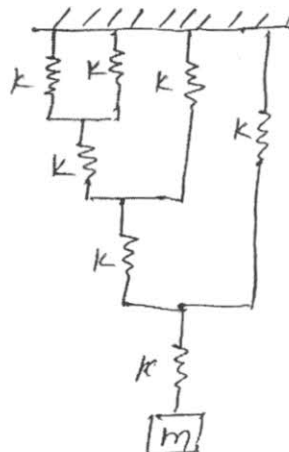


Fig. 8(a)

10 L3 5 3

- b) Determine the torsional natural frequency of the system shown in Fig. 8(b). Neglect the mass moment of inertia of the shaft. Take  $J=0.0098\text{kg-m}^2$

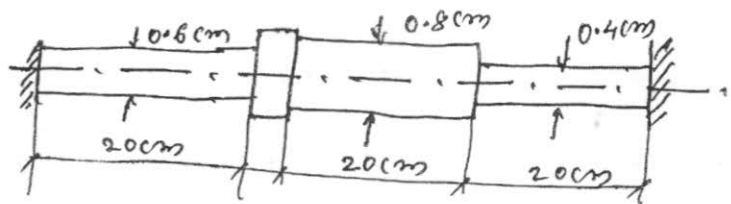


Fig. 8(b)

10 L3 4 2

## Unit - V

- |     |    |                                                                                                                                                                                                                               |    |    |   |   |
|-----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 9.  | a) | Explain different types of damping.                                                                                                                                                                                           | 6  | L1 | 5 | 1 |
|     | b) | Explain the following. i) Critical damping ii) Damping ratio iii) logarithmic decrement iv) damped natural frequency                                                                                                          | 6  | L1 | 5 | 1 |
|     | c) | Set up the differential equation for a spring mass-damper system and obtain complete solution for the critically damped condition.                                                                                            | 8  | L2 | 5 | 1 |
| 10. | a) | Derive an expression for logarithmic decrement.                                                                                                                                                                               | 8  | L1 | 5 | 1 |
|     | b) | In a vibrating system 100 kg mass supported by 100000 N/m spring constant with a damping constant of 632.4 N-s/m. Find the i) Natural frequency ii) Damped Natural frequency iii) Damped Ratio iv) Time period of oscillation | 12 | L3 | 5 | 2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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